



Zero-shot Sim2Real Wildfire Frontline Estimation from UAV Imagery via VLM-Guided Learning

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Background

Importance of Frontline in UAV-based Wildfire Analysis

- The **frontline** is the boundary where flames contact the ground, representing the active wildfire **spread boundary** in UAV imagery.
- Frontline information enables **accurate analysis of wildfire** progression and the actual fire extent from UAV imagery.

Challenges in Building Frontline Estimation Models

- Real wildfire imagery captured from UAV viewpoints is **difficult to acquire** due to the dangerous conditions of wildfire environments.
- Various UAV viewing conditions can make frontlines difficult to observe, complicating accurate and consistent **annotation**.

Synthetic Data Learning and the Sim2Real Domain Gap

- Simulation-based synthetic data** can generate UAV wildfire imagery with accurate frontline annotations, reducing reliance **on real-world data**.
- However, the **domain gap** between synthetic and real UAV wildfire imagery makes it difficult for models trained on synthetic data to generalize to real-world scenes.



Objectives

- Synthetic Wildfire Data Generation:** Develop a simulation-based synthetic data generation framework that enables models to learn frontline structures across **diverse wildfire environments**.
- VLM-guided Zero-shot Sim2Real Frontline Estimation:** Develop a VLM-guided approach that leverages semantic priors to mitigate the Sim2Real domain gap and enable **zero-shot frontline estimation** on real UAV wildfire imagery.



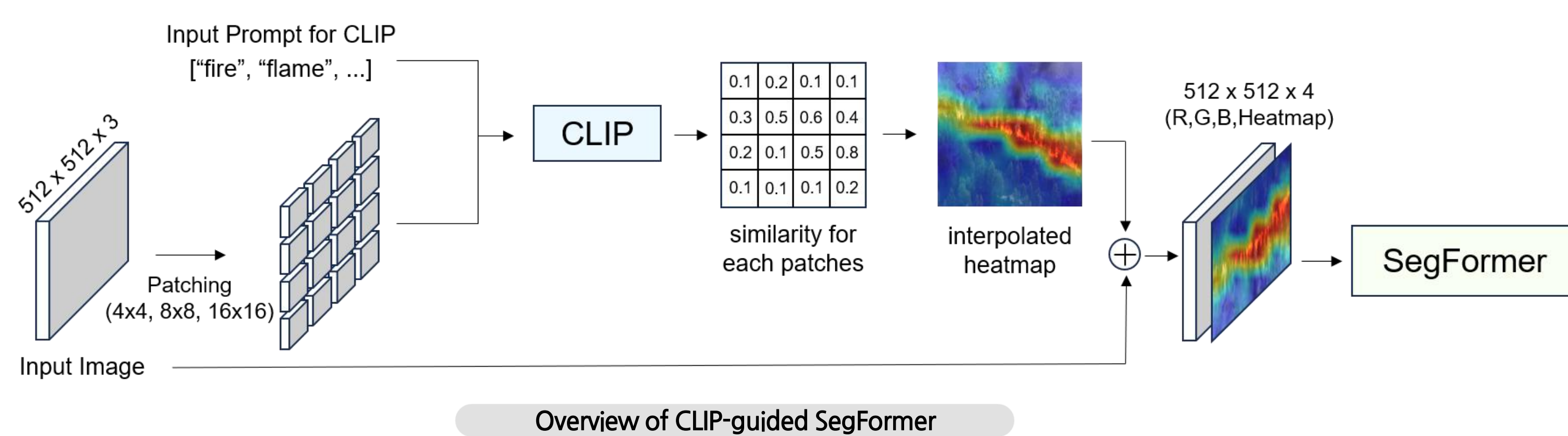
Methods

- Simulation-based Data Generation for Wildfire Frontline Learning

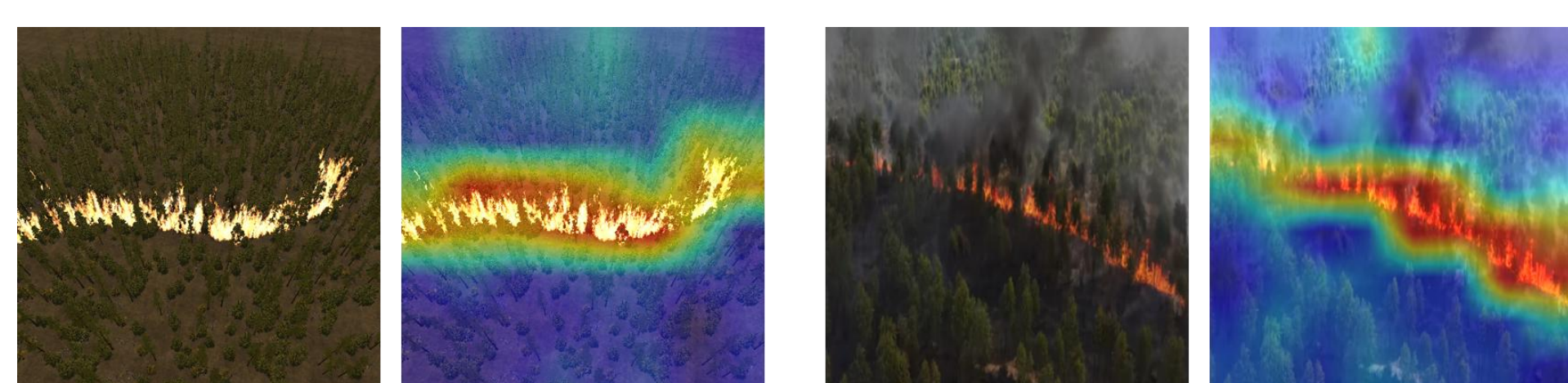


Synthetic wildfire with various randomization

- Generate diverse UAV wildfire scenes under **varied environmental conditions** while automatically producing **consistent frontline annotations**.
- CLIP-guided SegFormer for Sim2Real Frontline inference**



Overview of CLIP-guided SegFormer



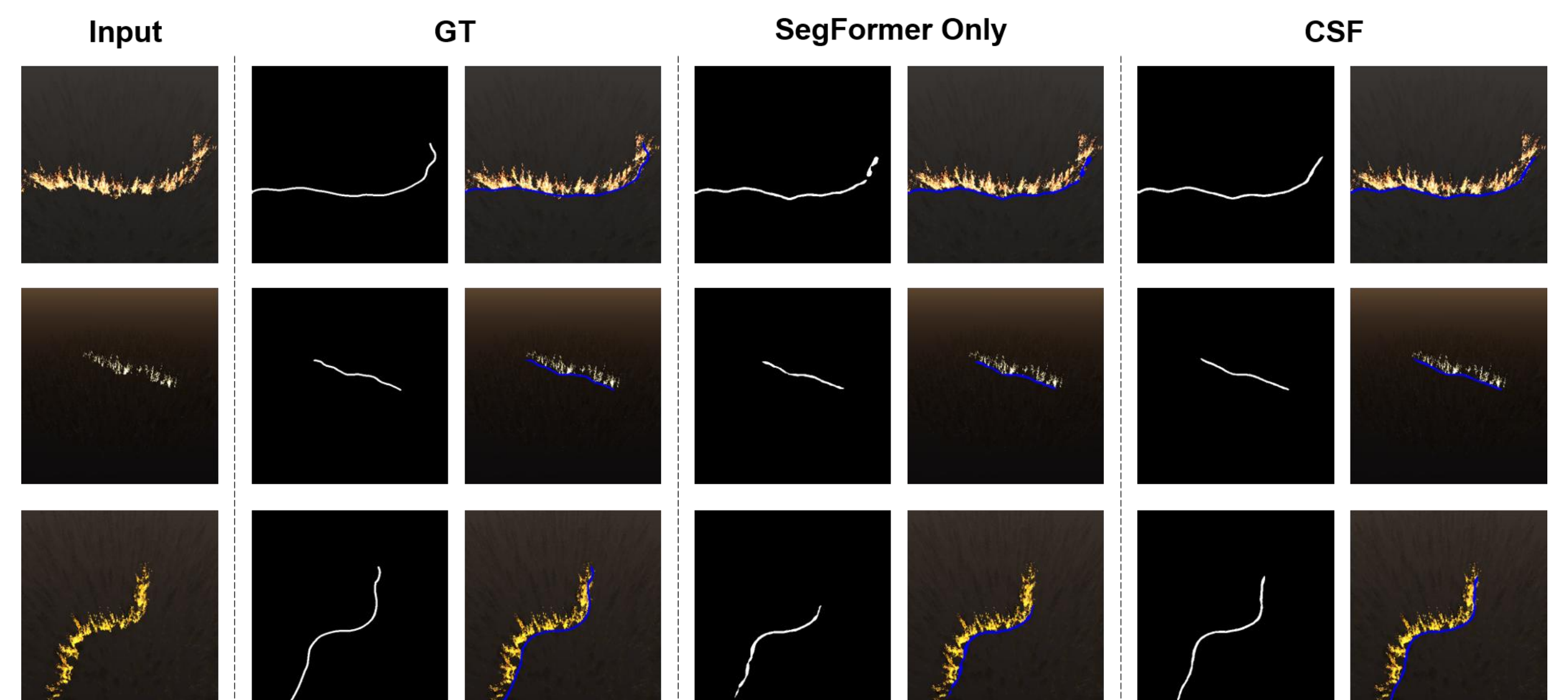
CLIP semantic heatmaps on synthetic and real imagery

- Generate **CLIP-based semantic heatmaps** from fire-related text prompts and concatenate them with RGB images as input to SegFormer.
- Guide the model to learn features centered on **semantically relevant fire regions**, mitigating the Sim2Real domain gap.



Results

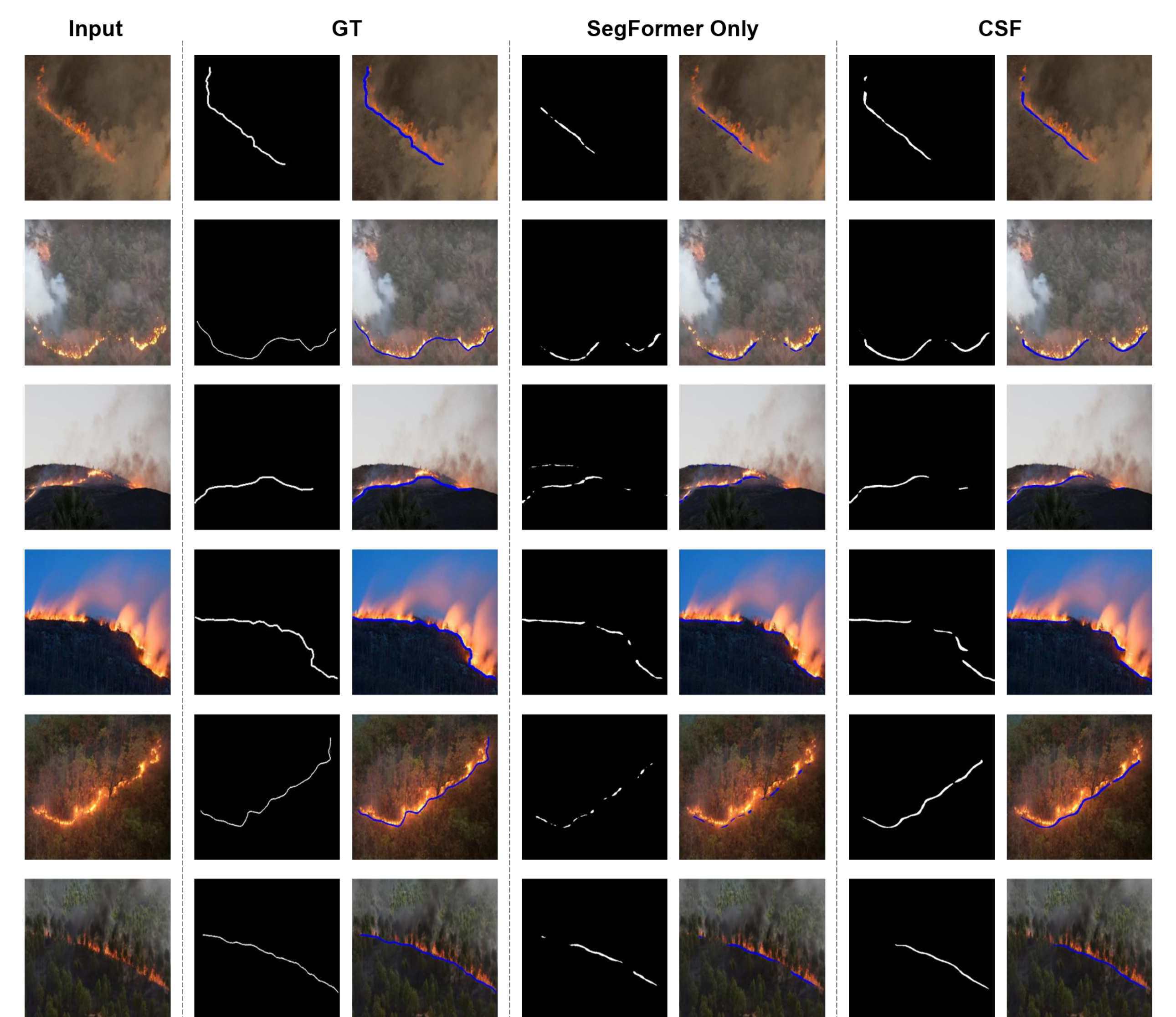
Comparison on Synthetic Data



Configuration	Precision	Recall	F1	Tol Pre.	Tol Rec.	Tol F1	ASSD
SegFormer	0.649	0.723	0.683	0.949	0.930	0.939	1.28
CSF	0.691	0.751	0.719	0.964	0.959	0.961	0.91

- Both SegFormer and CSF show reliable frontline estimation on **synthetic test data**.
- This provides a basis for comparing Sim2Real generalization on real wildfire data.

Comparison on Real Data



Configuration	Precision	Recall	F1	Tol Pre.	Tol Rec.	Tol F1	ASSD
SegFormer	0.454	0.270	0.323	0.764	0.461	0.560	20.28
CSF	0.434	0.356	0.385	0.747	0.579	0.647	9.71

- The performance gap between SegFormer and CSF becomes more pronounced on real **wildfire data**.
- CSF better preserves global frontline continuity, demonstrating improved **Sim2Real generalization**.
- These results indicate that CLIP-based semantic information **effectively mitigates the Sim2Real domain gap**.



Conclusions

- Proposes a **zero-shot Sim2Real** framework for wildfire frontline inference using only synthetic training data
- Combines synthetic data learning with **CLIP-based VLM guidance** to mitigate the **Sim2Real domain gap**.
- Future work will extend the framework to frontline tracking and fire region estimation.